Reaction Rates and Rate Laws

Answer key with worked solutions

A. Understanding

1. Collision theory.

> (a) Powder has a much **larger surface area**, so more particles are exposed to the acid and **collisions happen more often**. (b) A higher concentration means **more particles per litre**, so collisions are **more frequent**; the fraction of collisions with enough energy is unchanged, but there are more of them per second.

B. Rates and stoichiometry

2. Rates of formation.

> NO_2: 0.0060 M/s × (4 mol NO_2 / 2 mol N_2O_5) = **0.012 M/s**. O_2: 0.0060 M/s × (1 mol O_2 / 2 mol N_2O_5) = **0.0030 M/s**.

3. Average rate.

> $$-\frac{\Delta[\mathrm{A}]}{\Delta t} = -\frac{0.560\ \mathrm{M} - 0.800\ \mathrm{M}}{120\ \mathrm{s}} = \frac{0.240\ \mathrm{M}}{120\ \mathrm{s}} = 2.00 \times 10^{-3}\ \mathbf{M/s}$$

C. Rate laws

4. rate = $k[NO]^2[H_2]$.

> (a) **second order** in NO, **first order** in H_2 (b) **third order** overall (c) k = rate ÷ ($[NO]^2[H_2]$) = M s^{-1} ÷ M^3 = $\mathbf{M^{-2}\ s^{-1}}$ (d) the rate is multiplied by 2^2 = **4**

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5. Method of initial rates.

> Experiments 1 → 2: [NO] doubles, [O₂] constant; rate ratio = $\frac{1.12 \times 10^{-5}\ \text{M/s}}{2.8 \times 10^{-6}\ \text{M/s}} = 4$, so **second order in NO**.
>
> Experiments 1 → 3: [O₂] doubles; rate ratio = $\frac{5.6 \times 10^{-6}\ \text{M/s}}{2.8 \times 10^{-6}\ \text{M/s}} = 2$, so **first order in O₂**.
>
> Rate law: **rate = k[NO]²[O₂]**. From experiment 1:
>
> $$k = \frac{2.8 \times 10^{-6}\ \text{M s}^{-1}}{(0.020\ \text{M})^2(0.010\ \text{M})} = 0.70\ \text{M}^{-2}\ \text{s}^{-1}$$

6. Double [NO], halve [O₂].

> Rate is multiplied by $2^2 \times ½ = \mathbf{2}$: the rate doubles.

7. Units of k.

> Zero order: **M s⁻¹**; first order: **s⁻¹**; second order: **M⁻¹ s⁻¹**

D. Half-life

8. k = $2.31 \times 10^{-3}\ \text{s}^{-1}$.

> (a) $t_{1/2} = \frac{0.693}{2.31 \times 10^{-3}\ \text{s}^{-1}} = \mathbf{300.\ s}$
>
> (b) 900. s ÷ 300. s = 3 half-lives, so $[\text{A}] = 0.400\ \text{M} \times \left(\frac{1}{2}\right)^3 = \mathbf{0.0500\ M}$

E. Challenge

9. k = $0.0150\ \text{min}^{-1}$, $[\text{A}]_0 = 0.200\ \text{M}$.

> (a) $kt = 0.0150\ \text{min}^{-1} \times 60.0\ \text{min} = 0.900$ (no unit). $[\text{A}] = 0.200\ \text{M} \times e^{-0.900} = \mathbf{0.0813\ M}$
>
> (b) From $\ln\frac{[\text{A}]_0}{[\text{A}]_t} = kt$:
>
> $$t = \frac{\ln(0.200\ \text{M}/0.0500\ \text{M})}{0.0150\ \text{min}^{-1}} = \frac{1.386}{0.0150\ \text{min}^{-1}} = 92.4\ \text{min}$$
>
> (The concentration falls to a quarter, which is exactly 2 half-lives: 2 × 46.2 min = 92.4 min.)

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